

Assignment: Smart Task Management System

Objective

Build a simple Python-based web application to evaluate your practical skills in:

- Python
- Flask
- REST API Development
- PostgreSQL
- Pandas & NumPy
- WebSockets
- HTML/CSS

Project Requirements

Create a **Task Management System** with the following features:

1. Authentication

Using Flask:

- User Registration
- User Login
- Logout Functionality

2. REST API Development

Create APIs for:

- Add Task
- Update Task
- Delete Task
- Get All Tasks

Each task must contain:

- Title
- Description
- Priority
- Status

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- Created Date
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3. PostgreSQL Integration

Use PostgreSQL to store:

- User Data
- Task Data

Database structure should be properly organized.

4. Analytics Module

Using Pandas & NumPy, display:

- Total Tasks
 - Completed Tasks
 - Pending Tasks
 - Completion Percentage
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5. WebSocket Feature

Implement WebSockets for:

- Live task updates
OR
 - Real-time notifications
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6. Basic Frontend

Using HTML & CSS:

- Display task list
- Add tasks
- Show analytics summary

UI should be clean and responsive.

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Technical Requirements

Mandatory Technologies

- Python
- Flask
- PostgreSQL
- REST API
- Pandas & NumPy
- WebSockets

Evaluation Criteria

Criteria	Marks
Flask & REST APIs	25
PostgreSQL Integration	20
Code Quality	20
Pandas & NumPy Usage	15
WebSocket Feature	10
Frontend UI	10
Total: 100 Marks	

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Submission Requirements

Submit:

1. GitHub Repository Link
 2. Database File/Schema
 3. README.md with setup steps
 4. Demo Video (2–3 minutes)
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Important Notes

- Follow clean coding practices
 - APIs must work properly
 - Database integration is mandatory
 - Use proper project structure
 - Copied projects will be rejected
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Deadline

Complete and submit within **3–5 days**.

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